Chapter 9 — Structural Design Patterns (Study Edition)

> **Study Edition.** A shorter version of Chapter 9: every pattern and sub-topic is kept, the padding is removed. The full chapter (README.md) has every source file and full outputs.
>
> **Where this fits.** Chapter 8 was about *creating* objects; this chapter is about *putting objects together* — connecting, wrapping, nesting and sharing them. Chapters 10–11 cover how objects *talk* (behavioral patterns).

#	Pattern	One-line intent	Analogy
9.1	Adapter	Translate a class's interface into the one your code uses	Travel plug adapter
9.2	Bridge	Split two independently varying dimensions into two hierarchies joined by a reference	TV remote and TV sold separately
9.3	Composite	Treat one object and a tree of objects the same way	Folders hold files and folders
9.4	Decorator	Add behaviour to one object at run time by wrapping it	Coffee toppings
9.5	Facade	One simple entry point to a complicated subsystem	Hotel concierge
9.6	Flyweight	Share the repeating part of many objects to save memory	One glyph "e" reused on every page
9.7	Proxy	Same-interface stand-in that controls access	Receptionist guarding the manager
9.8	Comparison	Adapter vs Decorator vs Proxy vs Facade vs Bridge	–

Run: `cd chapters/09-patterns-structural/code && javac --release 21 -d /tmp/lld_09 $(find . -name '*.java') && java -cp /tmp/lld_09 lld.ch09.adapter.Main` (or `bridge`, `composite`, `decorator`, `facade`, `flyweight`, `proxy`). All seven compile with zero warnings under `-Xlint:all`; each top-level type has its own file and is package-private.

9.0 What structural patterns are

What it is

The GoF's 23 patterns fall into three families: **Creational** (Ch. 8 — how to create objects), **Structural** (this chapter — how to combine classes and objects into bigger structures) and **Behavioral** (Ch. 10–11 — how objects share work and talk). Structural patterns **simplify complex structures** (Facade, Composite), **promote reuse** (Adapter), **improve flexibility by decoupling** (Bridge, Decorator, Proxy) and **improve maintainability** (clear relationships, local changes). The seven GoF structural patterns: **Adapter, Bridge, Composite, Decorator, Facade, Flyweight, Proxy**.

Three words you need first

- **Composition** ("has-a") — an object holds and uses another; almost every structural pattern is built on it.
- **Delegation** — forwarding a call to the held object (`wrapper.pay()` calls `inner.charge()`).
- **Interface** (design sense) — the methods a caller can use on an object.

> **Key insight.** Adapter, Decorator, Proxy and Facade have the *same physical shape* — one object holds another and forwards calls. What separates them is **intent**: what changes as the call passes through (9.8).

9.1 Adapter Pattern

What it is

Adapter lets your code use a class whose methods have the "wrong" shape — one you can't change — through a small translator. GoF intent: *convert the interface of a class into another interface clients expect.* Also called **Wrapper**. Analogy: a travel plug adapter.

Role	Meaning	Ours
Target	Interface your code wants	`PaymentGateway`
Adaptee	Existing class with the wrong interface (usually not yours)	`StripeClient`
Adapter	Implements Target, translates to Adaptee	`StripeGateway`
Client	Uses the Target	`checkout`

The problem it solves

Checkout wants `pay(Money)`; the SDK offers `charge(amountInCents, currencyCode)` (or XML vs JSON, `track(String, Map)` vs your `Event`). **The other side isn't yours to change** — a vendor, another team, many existing callers. Stripe wants integer **cents** + **lowercase** currency; PayPal wants a **decimal string** "42.50" + **uppercase**; your code wants to think in `Money`.

How it works

1. **Target interface in your vocabulary** (`PaymentGateway.pay(Money) -> Receipt`), no vendor names.
2. **One adapter per adaptee**, implementing the Target and **holding** the vendor object.
3. **Translate inputs**, 4. **call the adaptee**, 5. **translate outputs and errors back** (vendor exception → your exception).
4. **Clients depend only on the Target.**

> **The rule:** the adapter's public methods mention **only your types** — parameters, returns *and* exceptions. Let a `StripeCardException` escape and callers depend on Stripe again.

Structure

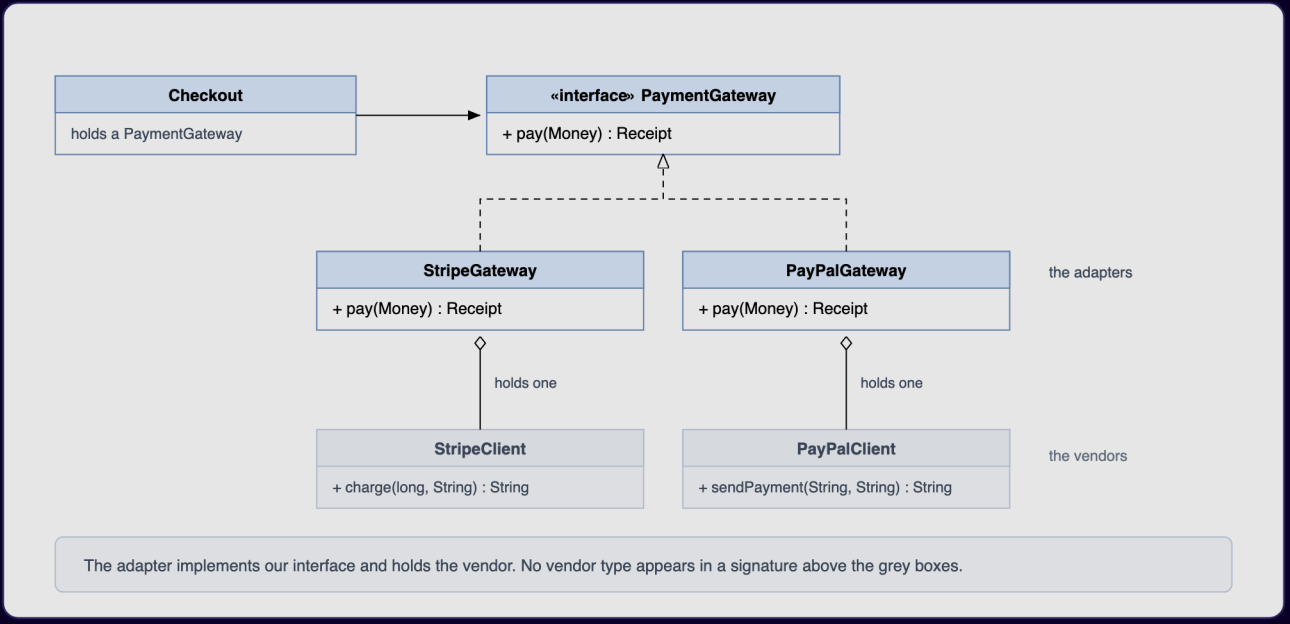
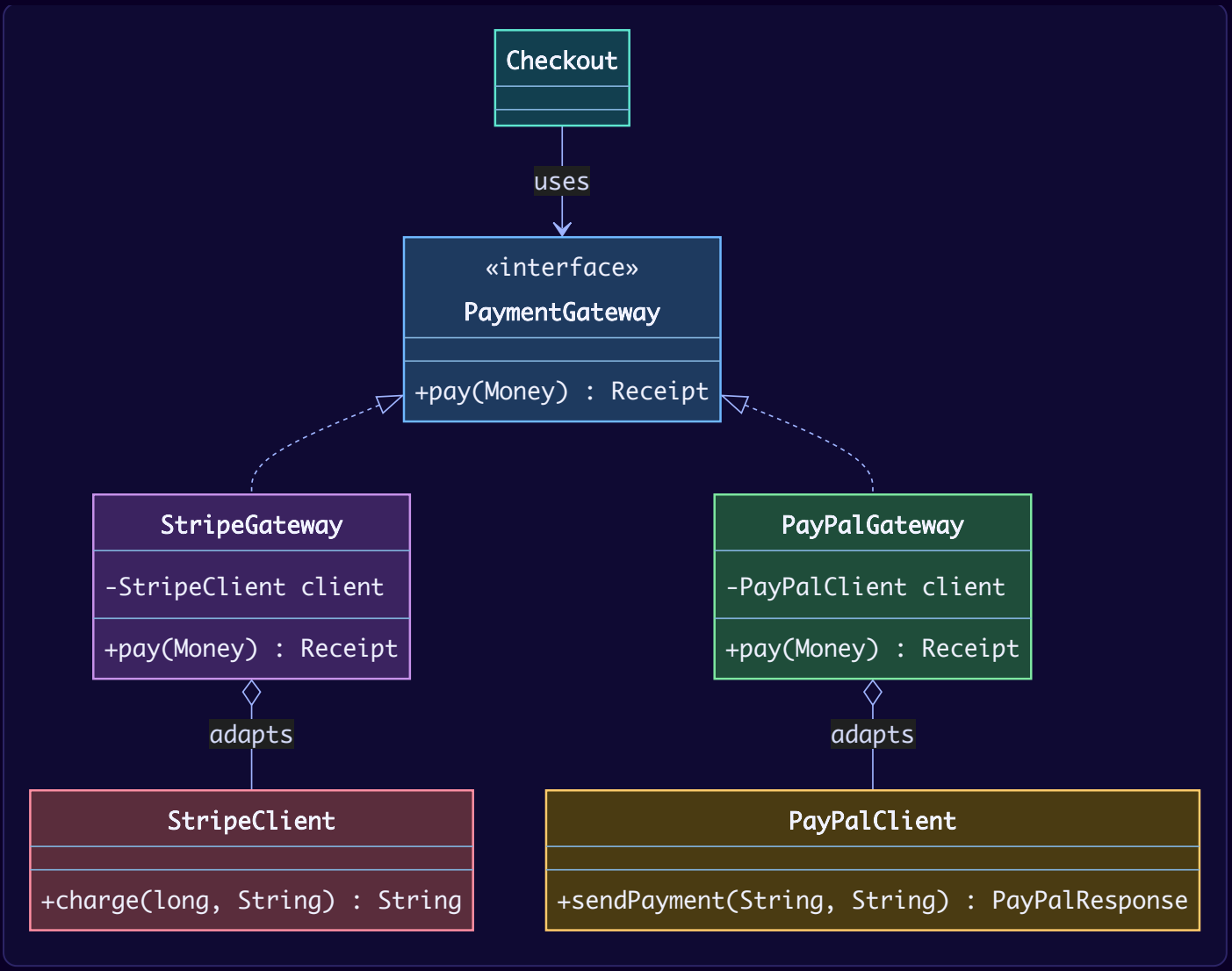


Figure — Adapter class diagram: `Checkout` holds a `PaymentGateway`; `StripeGateway` and `PayPalGateway` implement it and each wrap a vendor client, shown in grey because we do not own them (course diagram)



Java example

```java
/** TARGET interface: the shape our application wants to talk to. */
interface PaymentGateway {
    Receipt pay(Money amount);
}

/** OBJECT ADAPTER: implements our interface, HOLDS the vendor object (composition). */
final class StripeGateway implements PaymentGateway {
    private final StripeClient client;

    StripeGateway(StripeClient client) { this.client = client; }

    @Override
    public Receipt pay(Money amount) {
        try {
            String chargeId = client.charge(amount.cents(), amount.currency().toLowerCase(Locale.ROOT));
            return new Receipt(chargeId, "SUCCEEDED");
        } catch (StripeCardException e) {               // translate the vendor's error too
            throw new PaymentFailedException("Stripe declined: " + e.getMessage(), e);
        }
    }
}
```

`Money` (a record that upper-cases the currency once), `Receipt` and `PaymentFailedException` are *our* vocabulary. `PayPalGateway` does a different translation (`Money.asDecimal()` → "42.50"; `"approved"` → `"SUCCEEDED"`). `checkout(gateway, amount)` has no `if (stripe)` branch. The output shows `charge(4250, "usd")`,

`sendPayment("42.50", "USD")`, and a declined Stripe charge reaching the client as `payment failed -> Stripe declined: card_declined` – *our* error type.

Full runnable code: code/lld/ch09/adapter/ (see the Full Edition).

Object adapter vs class adapter

	Object adapter (composition)	Class adapter (inheritance)
How	*Holds* an adaptee, implements Target	*Extends* the adaptee, implements Target
Ours	`StripeGateway`, `PayPalGateway`	`ThermometerClassAdapter extends LegacyFahrenheitThermometer implements CelsiusSensor`
Adaptee `final`?	Works	**Fails** (our `StripeClient` is final)
Adapts adaptee + subclasses?	Yes	No – bound at compile time
Override adaptee behaviour?	Harder	Yes
Leaks adaptee API?	No	**Yes** (`readFahrenheit()` stays public)
Java constraint	none	Target must be an **interface** (no multiple class inheritance)

Prefer the object adapter (composition, works with `final`, no unwanted methods).

Two-way adapter

Implements **both** interfaces so clients of either side can use one object – `TwoWayTemperatureAdapter implements CelsiusSensor, FahrenheitSensor`: new code sees `100.0C`, old code `212.0F`. Handy during **migrations**.

Adapter vs Facade vs Decorator vs Proxy (quick view)

Pattern	Interface	Behaviour
Adapter	**Translated**	Same work, re-expressed
Decorator	Same	**Extended**
Proxy	Same	Same work, **gated** (defer, cache, refuse)
Facade	**Replaced** by one simpler one over many	Same work, fewer calls

Test: *adapters exist because someone else picked the interface* – if the vendor matched yours tomorrow, the adapter would be pointless. **Bridge** looks the same but is planned *before* the classes exist; an Adapter is bolted on *after*.

Trade-offs

Pros	Cons
Foreign format contained in one class	Extra class and call frame
Vendor change is local (Open/Closed)	**Conversions can drop information** silently
Easy to test – fake your one-method interface	Two data models must stay aligned
Your code keeps its language (SRP)	Logic (retries, pricing) creeps in

When not to use it

Both sides are yours (change one); the models truly disagree (redraw the boundary); you want extra behaviour (Decorator); the conversion happens in exactly one place.

Common mistakes

Leaking vendor types/exceptions; business logic in the adapter; class adapter by default; one adapter for many vendors with `if (vendor == STRIPE)`; silently mapping "pending" to "SUCCEEDED".

Real-world examples

`Arrays.asList` (array → `List`, no copy; writes go through, `add` throws); `InputStreamReader` / `OutputStreamWriter` (bytes ↔ chars); `Collections.enumeration`; SLF4J bindings; JDBC drivers; Spring MVC `HandlerAdapter`; every in-house "client" wrapping a vendor SDK.

In modern languages

A one-method Target makes a **lambda** the whole adapter (`FahrenheitSensor oldSensor = () -> 212.0;`); duck/structural typing (Python, TS, Go) makes adapters tiny.

In the interview

> **30-second answer:** "An Adapter converts one interface into another the client expects. I use it at boundaries I don't control — a vendor SDK, a legacy service. It implements *my* interface, holds the vendor object and translates inputs, outputs and exceptions, so no vendor type leaks in. Swapping vendors is one new class. I prefer the object adapter: composition, works with final classes."

- *Q: Adapter vs Decorator?* — A: Adapter changes the interface, keeps behaviour; Decorator keeps the interface, adds behaviour.
- *Q: Why not change the other side?* — A: Say it up front: it's a vendor library / another team / many callers. That constraint justifies the pattern.

Self-check

1. **Q:** The four roles? **A:** Target, Adaptee, Adapter, Client.
2. **Q:** Why can't `StripeClient` be class-adapted? **A:** It's `final`.
3. **Q:** `Arrays.asList(arr).add(x)`? **A:** `UnsupportedOperationException` — a fixed-size view over the array.

TL;DR

Translator between the interface you have and the one you need; implements **yours**, holds the vendor's, translates **in, out and errors**. Prefer object adapters. Only at boundaries you don't control.

9.2 Bridge Pattern

What it is

Bridge splits a class that varies in *two independent ways* into **two hierarchies** joined by a reference (the "bridge"). GoF intent: *decouple an abstraction from its implementation so the two can vary independently* (a.k.a. **Handle/Body**). Analogy: any remote works any TV via a common protocol.

Term	Meaning	Ours
Abstraction	High-level side, holds an Implementor	`Notification`
Refined Abstraction	Its subclasses	`AlertNotification`, `ReminderNotification`, `DigestNotification`
Implementor	Interface of the low-level physical work	`Channel`
Concrete Implementor	Real low-level work	`EmailChannel`, `SmsChannel`, `PushChannel`

> **Naming trap.** "Implementation" here isn't "a class implementing an interface" — it's the **low-level half** of the split (sending bytes, drawing pixels).

The problem it solves — the class explosion

Three notification types × three channels in one hierarchy = **9** classes (`EmailAlert`, `SmsDigest`…); add one of each → 16. With a bridge they **add**: 3 + 3 = 6, then 8. **Smell:** class names made of two nouns, each duplicating build or send code. Bonus: the pairing becomes a **run-time** choice (SMS in tests, email in production). The hard part is choosing which side is the abstraction.

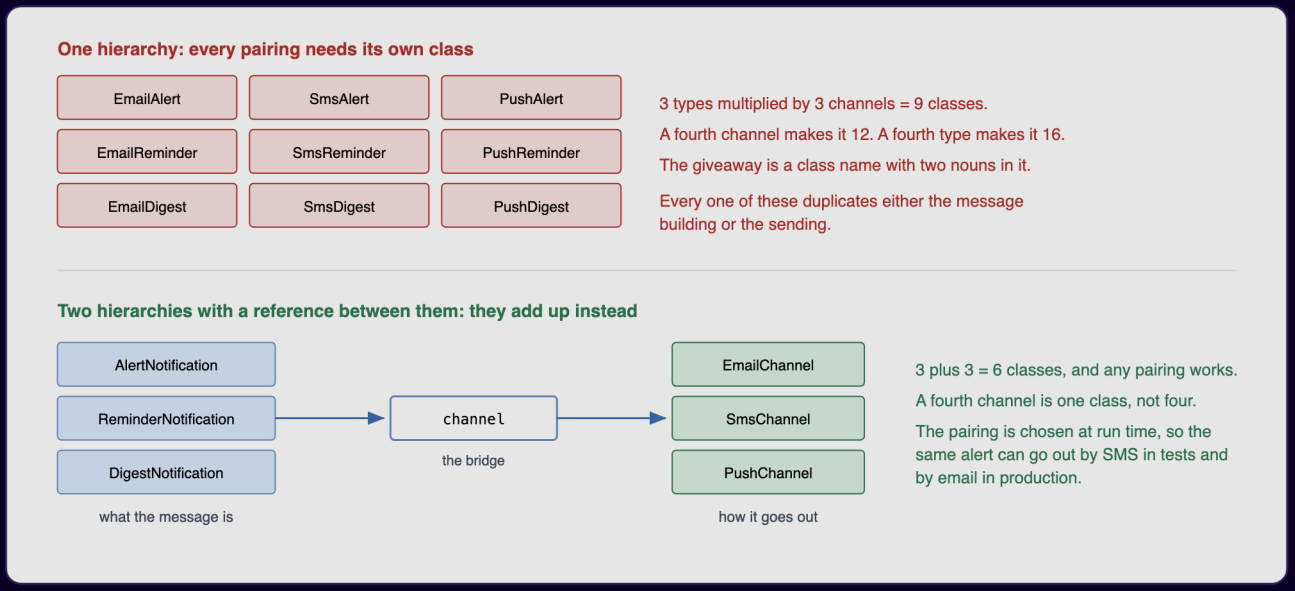
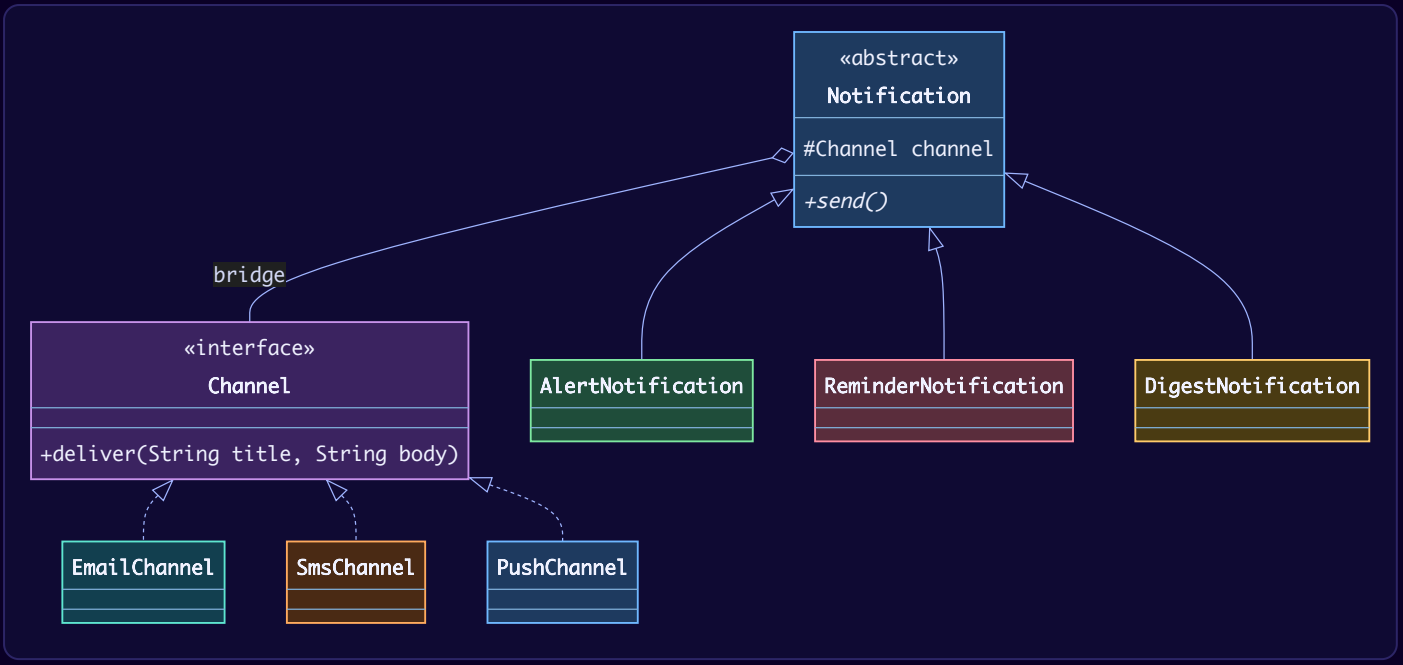


Figure — Top: one hierarchy needs 3 × 3 = 9 classes such as EmailAlert and SmsDigest. Bottom: two hierarchies joined by one `channel` reference need 3 + 3 = 6 (course diagram)

How it works

1. Find the two independent dimensions (what the message *is* vs how it *goes out*).
2. The lower-level one becomes a small, general **Implementor interface** (`Channel.deliver(title, body)`).
3. The higher-level one becomes an **abstract class holding a reference** to it — that field is the bridge.
4. Refined abstractions build content and **delegate** across the bridge, never checking the implementor type.
5. Pair at run time: `new AlertNotification(new SmsChannel(...), ...)`.

Structure



Java example

```
interface Channel {
    void deliver(String title, String body);
}

abstract class Notification {
    protected final Channel channel;

    protected Notification(Channel channel) { this.channel = channel; }

    /** Builds the message, then hands it across the bridge. Never branches on channel type. */
    abstract void send();
}

/** REFINED ABSTRACTIONS */
final class AlertNotification extends Notification {
    private final String system;
    private final String problem;

    AlertNotification(Channel channel, String system, String problem) {
        super(channel);
        this.system = system;
        this.problem = problem;
    }

    @Override
    void send() { channel.deliver("ALERT " + system, problem + " - act now"); }
}
```

`SmsChannel` truncates long text — an SMS rule living inside SMS only. Adding `SlackChannel` or `PromoNotification` = one class each.

Full runnable code: code/lld/ch09/bridge/ (see the Full Edition).

Bridge vs Strategy

Code is **identical** (hold an interface, delegate); the *reason* differs.

	Bridge	Strategy
What varies	A whole **dimension**	**One algorithm**
When chosen	Once, at construction	Often swapped, even per call
Why	Stop two hierarchies multiplying	Make one behaviour replaceable
Decided	Up front, structurally	Later, when a 2nd algorithm appears
Example	`Notification` → `Channel`	`list.sort(comparator)`

Test: a new implementation that is *a new kind of thing the system supports* → Bridge; *one more way to do one step* → Strategy.

Getting the split wrong

- **Low-level details leaking up** – `if (channel instanceof SmsChannel)` in `AlertNotification`; move it into `SmsChannel`.
- **An implementor shaped around one implementation** – `setEmailReplyTo(...)` on `Channel` forces others to throw. Keep it minimal (two plain strings).

Trade-offs

Pros	Cons
Classes add, not multiply	Two hierarchies to understand
Each side extends independently (Open/Closed)	More indirection
Pairing chosen at run time	**Hard to undo** once both sides grow
Platform code isolated (SRP)	Pointless with one implementor

When not to use it

An axis has one member (add it when the second lands); axes aren't independent (forbidden combos become run-time checks); you can't say in one sentence which side is the abstraction; the second axis is a behaviour (Strategy).

Common mistakes

Bridge "just in case"; abstraction branching on implementor type; implementor mirroring one implementation; confusing it with Adapter (retrofitted vs designed in).

Real-world examples

JDBC (`Connection` / `Statement` + drivers – also called Adapter); SLF4J API vs backend; GoF's `Window` / `WindowImp` per platform and graphics drivers; `java.awt` peers; `NotificationService` with SES/Twilio/FCM providers; DI binding per environment.

In modern languages

A one-method implementor can be a plain function; DI containers wire the pairing per environment.

Other patterns it is confused with

Looks like	Difference
Strategy	Same shape; permanent dimension vs swappable algorithm
Adapter	Retrofitted after vs planned before
Abstract Factory	Decides *which family is created*; often supplies the implementor a bridge receives
Decorator	Shares the wrapped interface and adds behaviour; Bridge sides have unrelated interfaces

In the interview

> **30-second answer:** "Bridge separates an abstraction from its implementation so both vary independently. Notifications by type and by channel in one hierarchy need 3 × 3 = 9 classes, 16 with a fourth of each. I split it: a `Notification` hierarchy holding a `Channel` interface – 6, then 8 – and the pairing is chosen at run time. I wouldn't build it with only one channel."

- *Q: vs Strategy?* – A: Same code; Bridge splits a permanent dimension up front, Strategy makes one algorithm replaceable.
- *Q: Who picks the implementor?* – A: Whoever builds it – often a factory or DI container, per environment.

Self-check

1. **Q:** 5 types × 4 channels, without/with Bridge? **A:** 20 vs 9.
2. **Q:** Which field is the bridge? **A:** `Notification`'s `Channel channel`.
3. **Q:** Bridge added before or after classes exist? **A:** Before (Adapter is after).

TL;DR

Two varying dimensions → two hierarchies + one reference; multiplication becomes addition. Keep the implementor small. Same code as Strategy, different intent.

9.3 Composite Pattern

What it is

Composite gives a single item and a nested group **one common interface**, so clients never check whether they hold a **leaf** or a **tree**. GoF intent: compose objects into **part-whole hierarchies** and treat individual objects and compositions **uniformly**. Analogy: ask any manager or individual contributor "your team's total salary?" – same question for both.

Term	Meaning	Ours
Component	Shared interface	`FileSystemNode`, `MenuComponent`
Leaf	No children; answers itself	`FileNode`, `MenuItem`
Composite	Holds children *typed as Component*, delegates	`FolderNode`, `Menu`
Client	Uses Components only	`report(...)`

The problem it solves

Nested data (folders, invoice bundles, UI panels) makes every operation start with `if (node instanceof Folder) ... else if (node instanceof File) ...`. That test is copied into `size`, `print`, `search`…; a new node type (symlink) must find every copy.

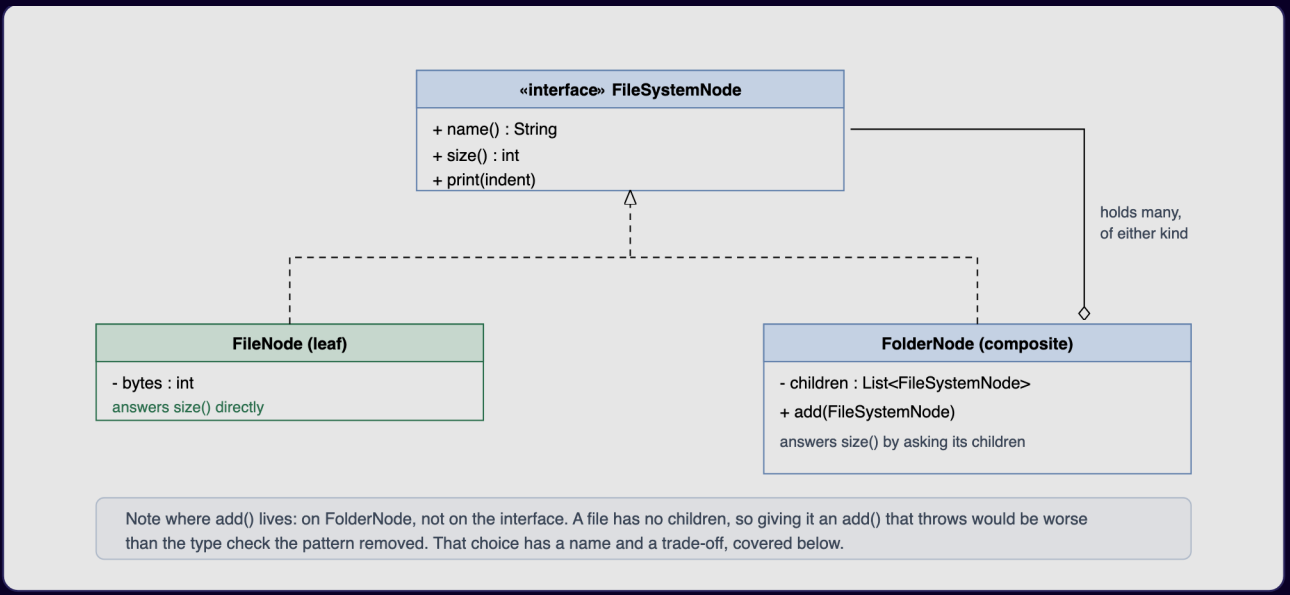
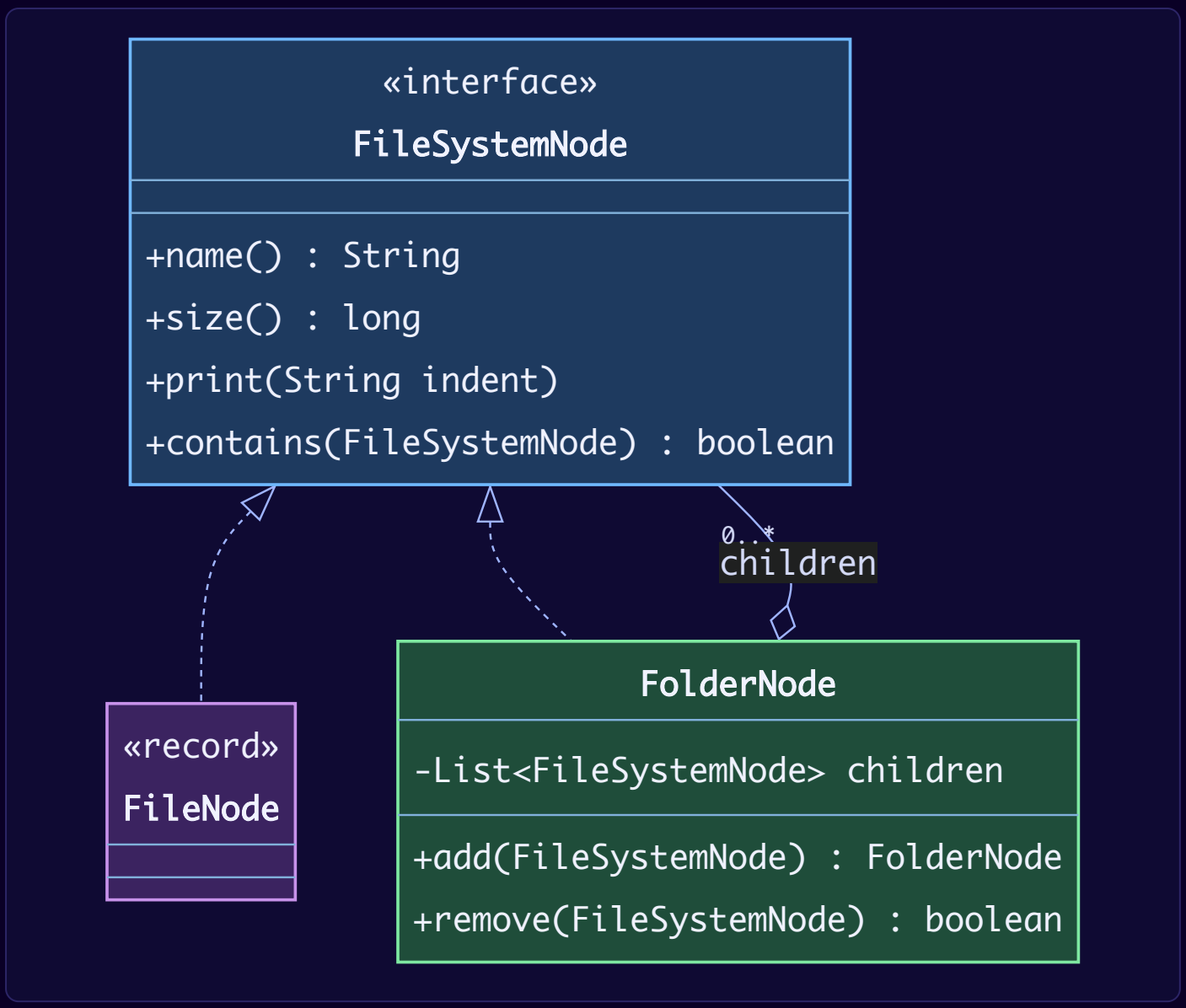


Figure — Composite class diagram: `FileNode` (leaf) and `FolderNode` (composite) implement `FileSystemNode`; the folder holds a list of `FileSystemNode` and has `add()` (course diagram)

How it works

1. A **Component** interface with operations valid for both (`name()`, `size()`, `print()`).
2. **Leaf** implements them directly.
3. **Composite** stores `List<Component>` — the **interface type** — and implements each operation by **asking children and combining**.
4. The recursion is written **once** and works to any depth.
5. **Clients** never type-check.

Structure


«interface»
FileSystemNode
+name() : String
+size() : long
+print(String indent)
+contains(FileSystemNode) : boolean
0..*
children
«record»
FileNode
FolderNode
-List<FileSystemNode> children
+add(FileSystemNode) : FolderNode
+remove(FileSystemNode) : boolean

Java example

```
interface FileSystemNode {
    String name();
    long size();
    void print(String indent);

    /** True if 'target' is this node or somewhere below it. Used for the cycle check. */
    default boolean contains(FileSystemNode target) { return this == target; }
}

/** LEAF: no children, answers directly. */
record FileNode(String name, long size) implements FileSystemNode {
    // ... print(indent) ...
}

/** COMPOSITE: holds children typed as the INTERFACE, so folders can hold folders. */
final class FolderNode implements FileSystemNode {
    private final String name;
    private final List<FileSystemNode> children = new ArrayList<>();
    // ... constructor ...
    /** Returns this so trees can be built fluently. Rejects cycles at insert time. */
    FolderNode add(FileSystemNode child) {
        if (child.contains(this)) {
            throw new IllegalArgumentException("cycle: " + child.name() + " already contains " + name);
        }
        children.add(child);
        return this;
    }
    // ... remove, children() returns Collections.unmodifiableList, name(), print, contains ...
    /** The recursion lives in ONE place. */
    @Override
    public long size() {
        long total = 0;
        for (FileSystemNode child : children) total += child.size();
        return total;
    }
}
```

`MenuComponent` / `Menu` / `MenuItem` is the *transparent* variant (*Head First*'s restaurant menu, prices in `BigDecimal`). Output: `report(project)` → 2150 bytes and `report(readme)` → 1200 with the same call; `src.add(project)` is rejected as a cycle; `Tea.add(...)` throws "Tea is a leaf".

Full runnable code: code/lld/ch09/composite/ (see the Full Edition).

Transparent vs safe composite

	Transparent	Safe
`add/remove` live	On the shared Component	Only on the Composite
Uniformity	Full, no casts	Builders must hold a `FolderNode`
Type safety	`leaf.add(...)` compiles, **throws at run time**	Compiler rejects `file.add(...)`
Ours	`MenuComponent` ("Tea is a leaf")	`FileSystemNode` / `FolderNode`
Choose when	Callers add children to arbitrary nodes (UI editor)	Tree built in one place, read in many (most cases)

Neither is wrong — check the choice was **deliberate**. *Head First* notes transparency knowingly bends SRP.

Two traps worth knowing

- **Cycles** – a folder inside its own descendant makes `size()` recurse forever (`StackOverflowError`). **Validate on insert.**
- **Repeated work** – `print()` calling `size()` at every level is ~quadratic; for big trees **cache subtree totals** and invalidate upward (needs parent pointers).

Trade-offs

Pros	Cons
No type checks in callers	Shared interface **bloats**
Recursion in one place	Weaker type safety (transparent)
New node types don't touch callers (Open/Closed)	Deep trees → deep stacks; cycles → endless recursion
Nesting for free	Totals recomputed unless cached

When not to use it

Nothing nests; leaves and containers share little (methods that throw are worse than the type test); deep trees on a hot path; trees modified during traversal.

Common mistakes

Child list typed as the leaf (`List<FileNode>`); returning the mutable child list (bypasses `add()`'s cycle check); `add()` on the interface by accident; no cycle check; using it where one recursive function with one type check would do.

Real-world examples

HTML DOM; `java.awt.Container`, Android `ViewGroup`; file systems; compiler ASTs; `java.io.File` (`listFiles()` returns `null` on a file – the transparent cost); JUnit 3 `TestSuite`; Spring `CompositeCacheManager`.

In modern languages

Sealed types + exhaustive `switch` (Java 21, Kotlin, Rust) are a real alternative – the compiler catches a forgotten node type; prefer them when the set of node types is closed. Functional languages: recursive ADTs + `fold`.

Other patterns it is confused with

Looks like	Difference
Decorator	Wraps **one** and adds behaviour; Composite holds **many** and combines
Iterator	Walks the tree without exposing it; often used together
Visitor	Adds new operations over the structure without editing nodes
Plain recursive function	Enough when only one operation needs the type check

In the interview

> **30-second answer:** "Composite lets a client treat a single object and a tree the same way. Leaf and container share one interface; the container stores children *typed as that interface* – which allows nesting – and delegates and combines. Its value is deleting the `instanceof` in every operation. I choose transparent or safe depending on whether callers need `add` on arbitrary nodes, reject cycles on insert, and cache totals for big trees."

- *Q: Transparent or safe?* – A: Safe if the tree is built in one place; transparent if callers add to arbitrary nodes.
- *Q: New operation without touching every node?* – A: Visitor.

Self-check

1. **Q:** Why `List<FileSystemNode>`, not `List<FileNode>`? **A:** So folders can contain folders.
2. **Q:** Transparent `add` on a leaf? **A:** Compiles, throws `UnsupportedOperationException`.
3. **Q:** Where to detect cycles? **A:** On insert.

TL;DR

Part-whole trees, one interface; leaf answers, composite asks children and combines. Child list holds the **interface**. Safe vs transparent; guard **cycles**; cache big totals.

9.4 Decorator Pattern

What it is

Decorator extends **one object** at run time by wrapping it in an object with **the same interface**; the original class is never edited. GoF intent: *attach additional responsibilities to an object dynamically – a flexible alternative to subclassing.* Also nicknamed **Wrapper**. Analogy: espresso + milk + mocha + whip – still a drink.

Term	Ours
Component	`UserRepository`, `Beverage`, `InputStream`
Concrete Component	`DatabaseUserRepository`, `Espresso`
Decorator (base) – holds a Component, forwards everything	`RepositoryDecorator`, `FilterInputStream`
Concrete Decorator – adds one behaviour	`LoggingRepository`, `CachingRepository`, `Mocha`, `UpperCaseInputStream`

Key property: **a decorator IS-A and HAS-A Component**, so it can wrap the real object, another decorator or a test double.

The problem it solves

Logging, caching, retries, metrics – each caller wants its own mix. **Subclassing**: 4 optional concerns → 2^4 = **16** classes, doubling per concern. **Flags**: `new Repo(true, false, true, true)`. Decorator: **one class per concern** (4), stacked at run time – Open/Closed.

How it works

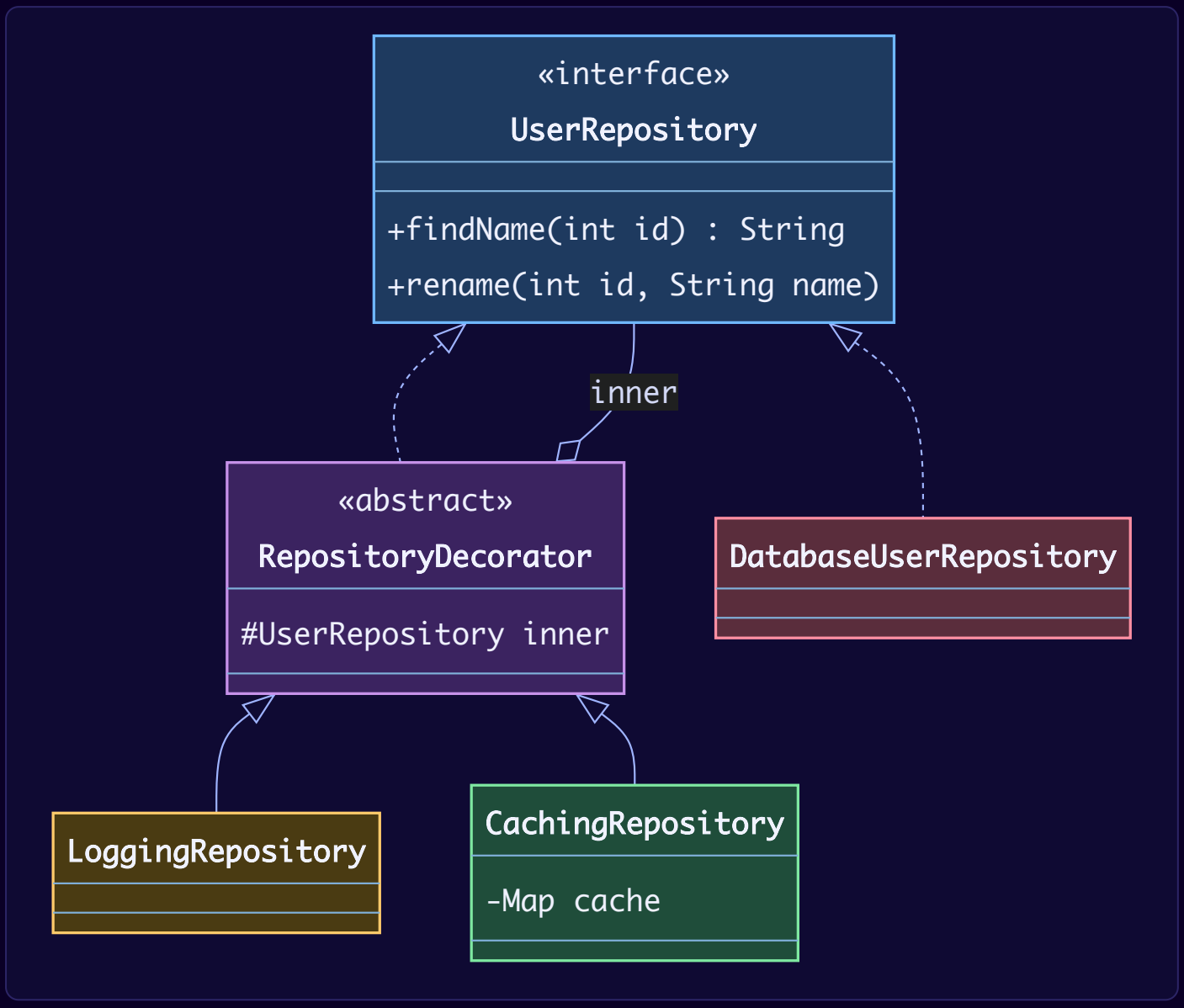
1. **Component interface**. 2. **Concrete component** doing the real work. 3. **Abstract base decorator** holding an `inner` of the **interface type**, **forwarding every method**. 4. **Concrete decorators** override only what they change, working before/after `inner`. 5. **Compose**: `new Logging(new Caching(new Database()))`.



Figure — The same three classes stacked in two orders. Logging outside caching logs every call; caching outside logging means the second call (a cache hit) is never logged (course diagram)

Structure

The bug that bites is **forgotten delegation** — a decorator that overrides `findName` but swallows `rename`. The abstract base decorator forwarding everything is the Java guard.



Java example

```
abstract class RepositoryDecorator implements UserRepository {
    protected final UserRepository inner;

    protected RepositoryDecorator(UserRepository inner) { this.inner = inner; }

    @Override public String findName(int id)          { return inner.findName(id); }
    @Override public void rename(int id, String name) { inner.rename(id, name); }
}

/** CONCRETE DECORATOR 2 - note it must also invalidate on writes. */
final class CachingRepository extends RepositoryDecorator {
    private final Map<Integer, String> cache = new HashMap<>();

    CachingRepository(UserRepository inner) { super(inner); }

    @Override
    public String findName(int id) {
        if (cache.containsKey(id)) {
            System.out.println("    [cache] hit for " + id);
            return cache.get(id);
        }
        System.out.println("    [cache] miss for " + id);
        String name = super.findName(id);
        cache.put(id, name);
        return name;
    }

    @Override
    public void rename(int id, String name) {
        cache.remove(id);
        super.rename(id, name);
    }
}

/** Add-ons are decorators: they ARE a Beverage and HAVE a Beverage. */
record Milk(Beverage inner) implements Beverage {
    public String description() { return inner.description() + ", Milk"; }
    public int costCents() { return inner.costCents() + 10; }
}
```

`LoggingRepository` prints then delegates. `new Whip(new Mocha(new Mocha(new Espresso())))` = 1.99 + 0.20 + 0.20 + 0.10 = **$2.49** (double mocha = wrap twice; prices in int cents). `UpperCaseInputStream extends FilterInputStream` is our own `java.io` decorator (ASCII only).

Full runnable code: code/lld/ch09/decorator/ (see the Full Edition).

Layer order changes behaviour

- **Logger outside, cache inside** – every request logged → the log shows *what callers asked*.
- **Cache outside, logger inside** – only misses logged → the log shows *what hit the database*.

Pair	Outside	Inside
Retry vs cache	Repeats the whole lookup	Repeats only the DB call; success fills the cache. Put retry **next to what fails**
Metrics vs cache	Caller-perceived time	DB-only time (what on-call is paged on); often record both
Authorization vs cache	**Correct**	**Security bug** – a cache hit skips the check

The compiler accepts every order – **comment the reason where the stack is built.**

Two related traps

- **Stateful decorators applied twice** (a counter counts every call twice).
- **Lost identity** – `==`, `equals`, `instanceof` on the wrapper differ from the wrapped object.

Decorator vs inheritance

	Subclassing	Decorator
Chosen	Compile time	Run time
How many	One subclass	Any number, any order
Classes for N features	Up to 2^N	N
`final` / not-owned classes	No	Yes (needs only the interface)
Private state	Protected members	No – public interface only

Java I/O streams: the decorator you already use

`new BufferedReader(new InputStreamReader(new FileInputStream("data.txt"), UTF_8))` – `FileInputStream` is the concrete component, `InputStreamReader` is strictly an **adapter** (bytes → chars), `BufferedReader` a **decorator**. `FilterInputStream` is the byte-side base decorator; `BufferedInputStream`, `DataInputStream`, `PushbackInputStream`, `GZIPInputStream`, `DigestInputStream` are concrete decorators. Downside (*Head First*): many small classes.

Trade-offs

Pros	Cons
Any mix without a class per mix (4, not 16)	Deeper stack traces
Concerns tested in isolation (SRP)	**Identity surprises**
Assembled at run time	**Order matters**, unchecked
Wrapped class never edited (Open/Closed)	Over-layering

When not to use it

Exactly one permanent concern; the concern needs private state; the framework already offers it (middleware, interceptors, Spring AOP); the stack would get deep.

Common mistakes

Holding the concrete class instead of the interface (can't stack); forgetting to forward methods; a cache that doesn't invalidate on writes; authorization inside a cache; relying on `instanceof` of the outer object.

Real-world examples

`java.io`; `Collections.unmodifiableList/synchronizedList/checkedList`; `HttpServletRequestWrapper`; HTTP middleware; Resilience4j/Polly retry/circuit-breaker stacks (order documented); Python `@lru_cache`.

In modern languages

Python function decorators, JS higher-order functions, Go struct embedding, C# `DelegatingHandler`.

In the interview

> **30-second answer:** "Decorator adds responsibilities at run time by wrapping an object in another with the same interface. Four optional concerns would be sixteen subclasses; with decorators each concern is one class and I compose the stack. The decorator holds the interface, which lets layers stack. Order matters: retry closest to what fails, authorization outside any cache."

- *Q: Decorator vs Proxy?* – A: Same structure; Decorator **adds behaviour** to an object it's handed (often stacked), Proxy **controls access** and often creates the real object itself. The boundary is fuzzy – a caching wrapper can be either.
- *Q: vs Chain of Responsibility?* – A: In Decorator every layer normally works and forwards; in CoR typically one handler handles the request.

Self-check

1. **Q:** 5 optional features: decorators vs subclasses? **A:** 5 vs up to 32.
2. **Q:** Why is "authorization inside cache" a bug? **A:** A cache hit returns data without the check.
3. **Q:** Is `InputStreamReader` a decorator? **A:** No – an adapter; `BufferedReader` is the decorator.

TL;DR

Wrap to add behaviour, keep the interface (IS-A + HAS-A). N features = N classes, any order – **order matters**. Java I/O, `synchronizedList`, middleware, Resilience4j. A decorator that short-circuits drifts toward Proxy.

9.5 Facade Pattern

What it is

Facade gives a complicated subsystem **one simple front door**. GoF intent: *provide a unified interface to a set of interfaces in a subsystem; a higher-level interface that makes it easier to use.* Analogy: a hotel concierge – you can still call the restaurant yourself.

Term	Ours
Facade – knows which classes to call, in what order, and the cleanup	`OrderFacade`
Subsystem classes – do the work, unaware of the facade	`InventoryService`, `PaymentService`, `ShippingService`, `EmailService`
Client	`Main` / checkout page

The problem it solves

Placing an order = **reserve stock → charge card → create shipment → send confirmation**, repeated by every caller – and each must also undo the reservation if the card is declined. The **undo** is where bugs live: forget it and units stay locked. The subsystem is fine (flexible = many knobs); callers just keep rewriting the standard path.

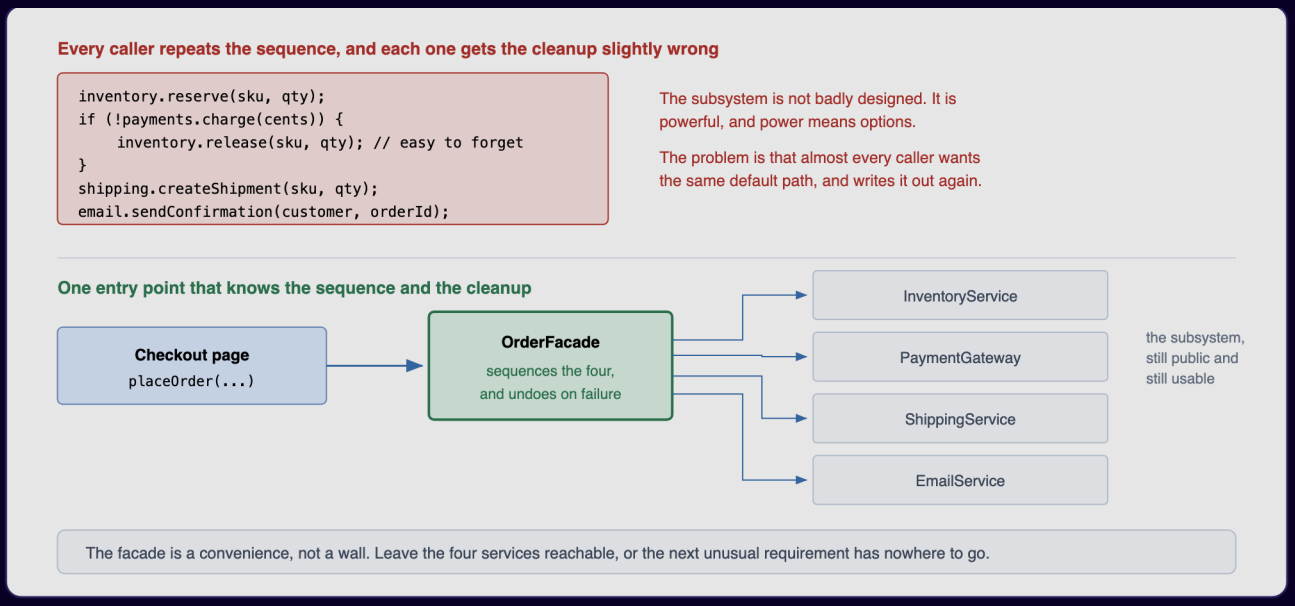


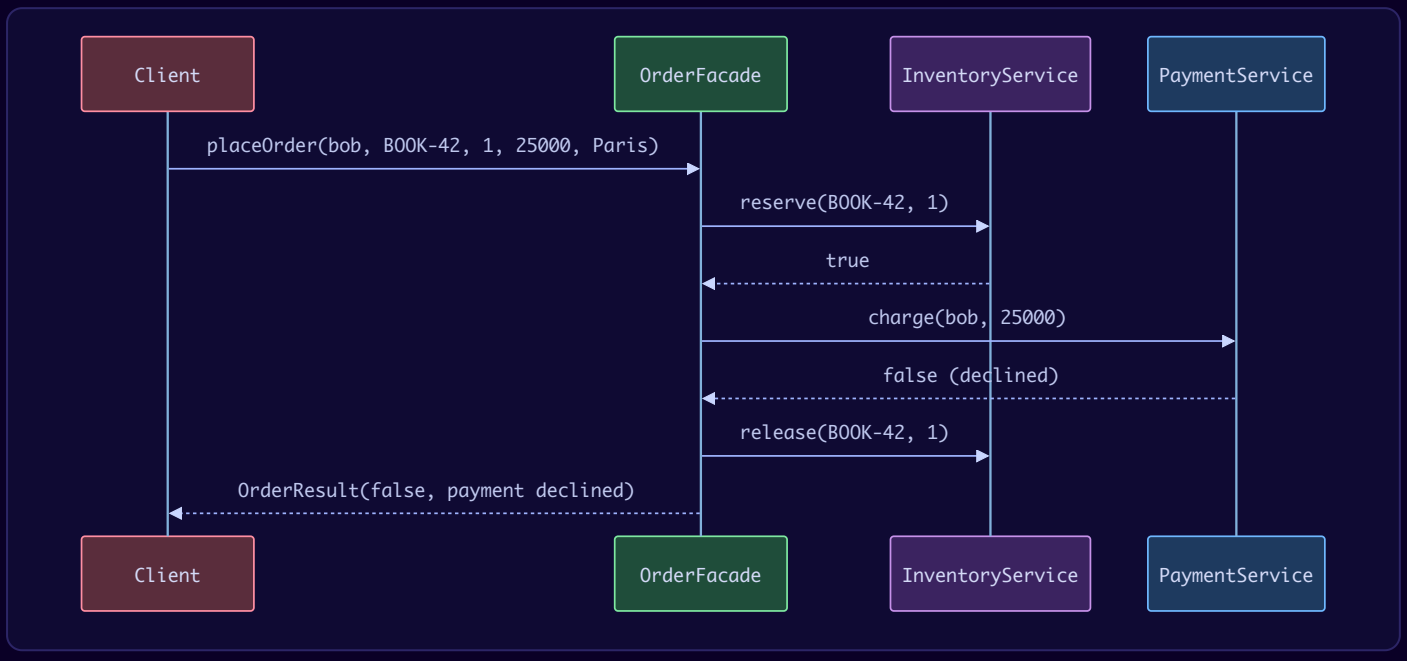
Figure — Top: each caller repeats the reserve/charge/release/ship/email sequence and may forget the release. Bottom: the checkout page calls `OrderFacade.placeOrder(...)`, which sequences the four services and undoes on failure; the services stay public (course diagram)

How it works

1. Identify the **common path**. 2. A **facade class** holding the subsystem objects. 3. One **intention-revealing method** (`placeOrder`) calling them in order. 4. **Own the failure handling** (a *compensating action*: release on decline). 5. **Keep the subsystem public**. 6. **No business rules** — it *sequences*, not *decides*.

Structure

Dependencies run **one way** — no service knows the facade. Usually a concrete class, no interface (unless you'll swap the whole subsystem).



Java example

```
final class OrderFacade {
    private final InventoryService inventory;
    private final PaymentService payments;
    private final ShippingService shipping;
    private final EmailService email;
    private int orderSeq = 100;
    // ... constructor takes the four services ...
    OrderResult placeOrder(String customer, String sku, int qty, long cents, String address) {
        String orderId = "ORD-" + orderSeq++;
        if (!inventory.reserve(sku, qty)) {
            return new OrderResult(false, orderId, "out of stock");
        }
        if (!payments.charge(customer, cents)) {
            inventory.release(sku, qty);            // compensating step callers used to forget
            return new OrderResult(false, orderId, "payment declined");
        }
        String shipmentId = shipping.createShipment(sku, qty, address);
        email.sendConfirmation(customer, orderId);
        return new OrderResult(true, orderId, "shipped as " + shipmentId);
    }
}
```

`OrderResult` is our own small result type (no subsystem types leak). `Main` still calls `inventory.reserve(...)` directly — **a convenience, not a wall**.

Full runnable code: code/lld/ch09/facade/ (see the Full Edition).

Facade vs Adapter vs Mediator

	Facade	Adapter	Proxy	Mediator (Ch. 10–11)
Intent	Simplify a subsystem	Convert an interface	Control access	Coordinate peers
Objects wrapped	**Several**	One	One	Several colleagues
Interface	New, smaller	Client's Target	**Same** as real	Mediator's own
Who chose it	**You**	**Someone else**	Fixed by real object	You
Wrapped objects know it?	**No**	No	No	**Yes** — talk through it
Direction	Client → facade → subsystem	Client → adapter → adaptee	Client → proxy → subject	Colleague ↔ mediator ↔ colleague
Example	`OrderFacade.placeOrder`	`StripeGateway.pay`	`LazyReport`	Air-traffic control; chat room

Fastest test: Adapter/Decorator/Proxy stand in front of **one** object; a Facade in front of **several**. If the subsystem's API became perfect tomorrow, a Facade would still simplify; an Adapter would be pointless. **Facade vs service layer:** same thing while it only sequences; with business rules it has become a domain/application service.

The failure mode to watch for

Facades **pull in features** — a discount rule, then a fraud check, then a **god object**. Rule: **a facade orchestrates; it does not decide.** Related principle (*Head First*): **Least Knowledge / Law of Demeter** — clients talk to the facade, not chains of subsystem objects.

Trade-offs

Pros	Cons
One call replaces a sequence	One more class to maintain
Order and undo in one place	**Feature magnet** → god object
One seam for swapping the subsystem	Opinionated — some caller needs an omitted option
Smaller public surface, lower coupling	Can mask problems; single point of failure

When not to use it

The subsystem is already simple; callers need the full range (every caller bypasses it); covering a mess instead of fixing it; wanting it to be the *only* entrance.

Common mistakes

Business rules in the facade; making the subsystem inaccessible; returning subsystem types; forgetting the failure path; confusing it with Mediator.

Real-world examples

Application service layers (`OrderService`); `java.net.URL.openStream()` (constructors deprecated since Java 20 — use `URI.create(s).toURL()`); SLF4J "Simple Logging Facade"; `FacesContext`; jQuery's `$`; Python `requests` over `urllib3`; e-commerce "Place order" orchestrators / backend-for-frontend.

In modern languages

Often just a module or package exporting one function (or a JPMS module exporting only the facade package).

In the interview

> **30-second answer:** "A Facade gives one simple entry point to a complex subsystem. `OrderFacade.placeOrder` reserves, charges, ships and emails in order, and releases the reservation if payment fails, so no caller has to remember the cleanup. It wraps several objects, unlike Adapter, Decorator and Proxy. I keep the subsystem public and business rules out — the risk is a god object."

- *Q: vs Mediator?* — A: Subsystem classes don't know the facade; colleagues know the mediator and talk through it both ways.
- *Q: Several facades?* — A: Yes — one per use case, or layered.

Self-check

1. **Q:** Fastest structural tell vs Adapter/Decorator/Proxy? **A:** It wraps several objects.
2. **Q:** What happens on a declined payment? **A:** It releases the reservation and returns a failed `OrderResult`.
3. **Q:** Add a 10% discount inside `placeOrder`? **A:** No — the facade sequences, a service decides.

TL;DR

One door into a complex subsystem; names the common path and owns the cleanup. Wraps **several**; subsystem stays **public**; **sequences, never decides**.

9.6 Flyweight Pattern

What it is

Flyweight shares the part of each object's state that is the same across many objects so huge numbers fit in memory. GoF intent: *use sharing to support large numbers of fine-grained objects efficiently.* Analogy: one type design for every "e", placed at different positions.

Term	Meaning	Map example
Intrinsic state	Identical for every holder; lives **inside** the shared flyweight	icon bitmap, colour, size
Extrinsic state	Varies per context; stays **outside**, **passed in** as arguments	latitude, longitude

Role	Ours
Flyweight (shared, immutable, intrinsic only)	`MarkerStyle`
Flyweight factory (owns the pool)	`MarkerStyleFactory`
Context (extrinsic state + reference)	`Marker`
Client	`Main`

GoF's own list: *Flyweight* (interface accepting extrinsic state), *ConcreteFlyweight* (`MarkerStyle` plays both), *UnsharedConcreteFlyweight* (same interface, not shared), *FlyweightFactory*, *Client*; "Context" is a later name (refactoring.guru).

The problem it solves

100,000 map markers, each with its own position plus a 2 MB icon bitmap → ~195 GB: the program **runs out of memory**. Same for a million game trees sharing one mesh, or font data per character.

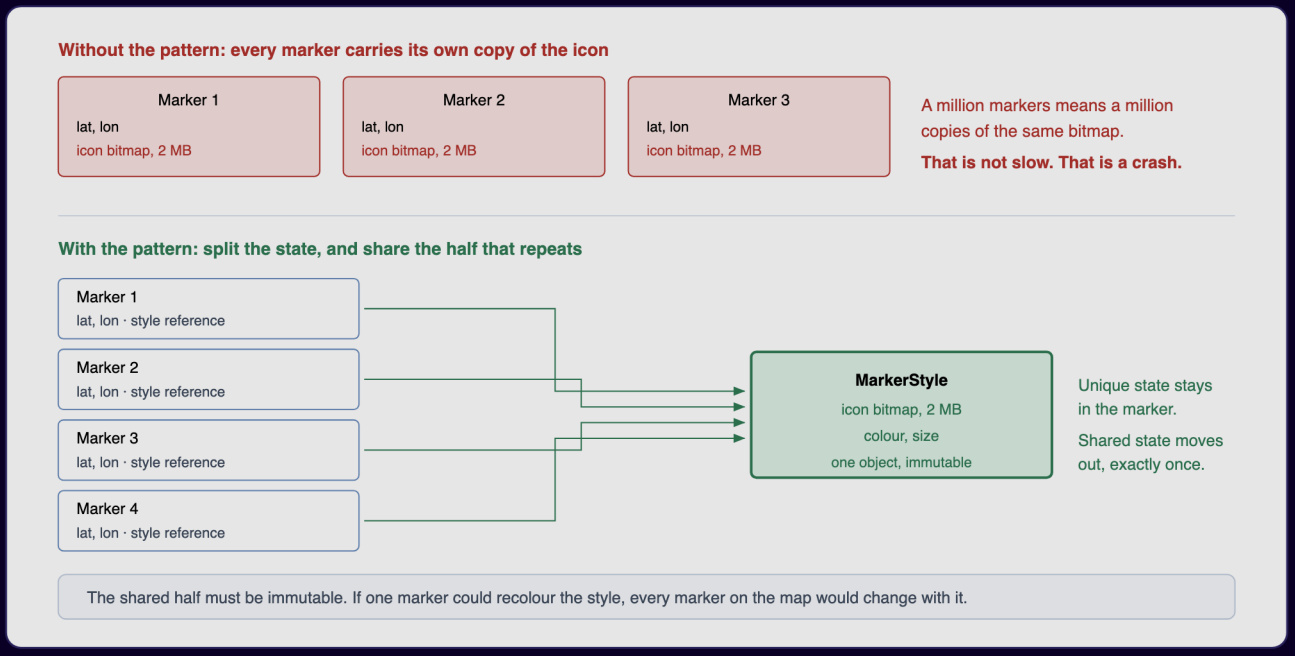


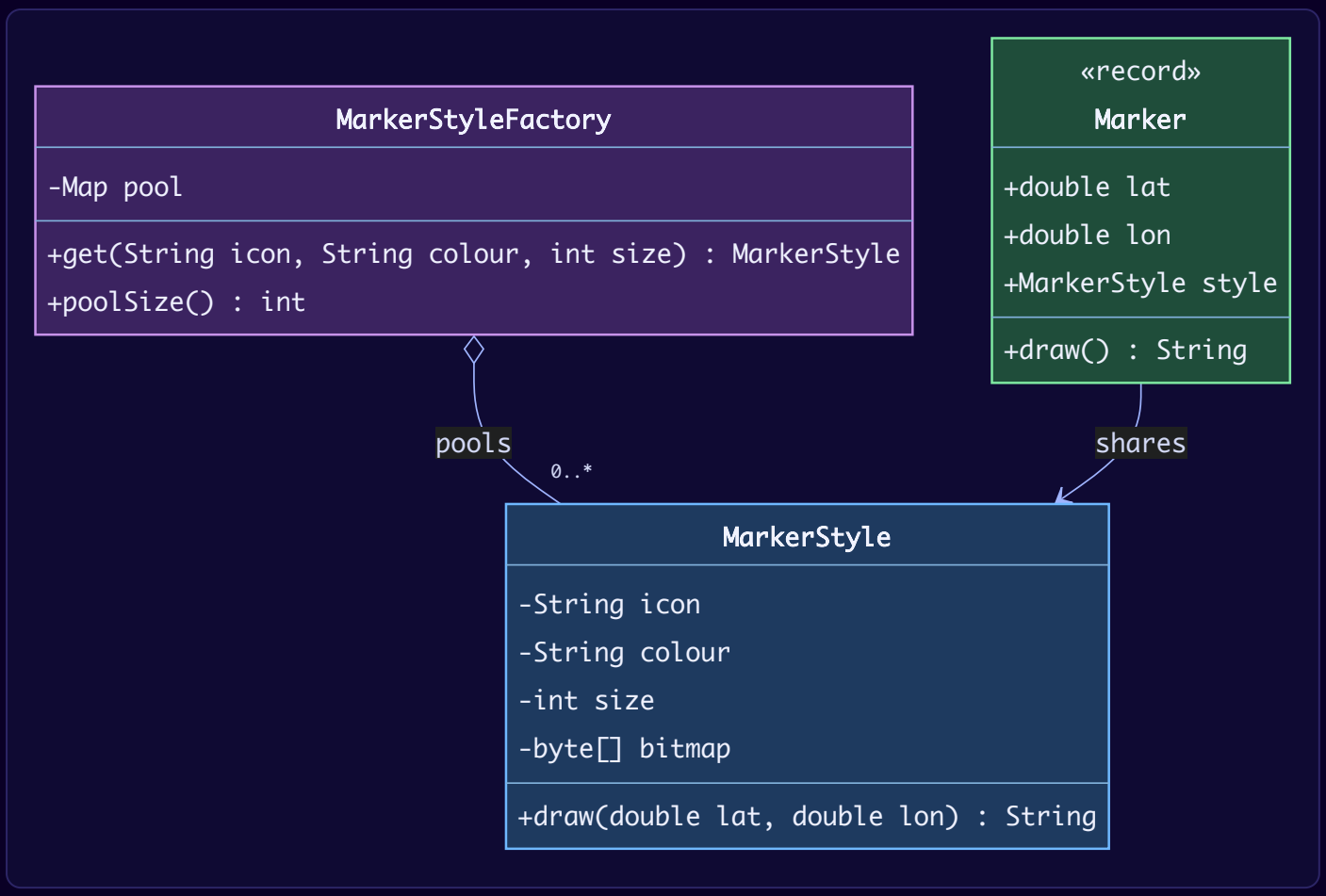
Figure — Top: every marker carries its own 2 MB icon. Bottom: each marker keeps only lat/lon and a style reference; one immutable `MarkerStyle` holds the icon, colour and size (course diagram)

How it works

1. **Split** intrinsic vs extrinsic. 2. Intrinsic → an **immutable** flyweight (`final`, no setters). 3. Extrinsic → **method arguments** (`draw(lat, lon)`). 4. A **factory owns the pool**, keyed by intrinsic values, returning existing or creating once. 5. Clients **only use the factory** (non-public constructor). 6. Contexts hold a **reference**, not a copy.

> **Two rules — break either and the pattern silently fails: (1) flyweights never change; (2) only the factory creates them.**

Structure


MarkerStyleFactory
-Map pool
+get(String icon, String colour, int size) : MarkerStyle
+poolSize() : int
«record»
Marker
+double lat
+double lon
+MarkerStyle style
+draw() : String
pools
0..*
shares
MarkerStyle
-String icon
-String colour
-int size
-byte[] bitmap
+draw(double lat, double lon) : String

Java example

```
final class MarkerStyle {
    static final int ICON_BYTES = 2 * 1024 * 1024;   // pretend every icon bitmap is 2 MB

    private final String icon;
    private final String colour;
    private final int size;
    private final byte[] bitmap;   // the heavy part we want exactly one copy of
    // ... package-private constructor loads the bitmap ...
    /** EXTRINSIC state (the position) is passed in, never stored. */
    String draw(double lat, double lon) {
        return String.format("%s %s marker at (%.4f, %.4f)", colour, icon, lat, lon);
    }
    // ... bitmapBytes() ...
}

/** FLYWEIGHT FACTORY: owns the pool. Thread-safe via ConcurrentHashMap.computeIfAbsent. */
final class MarkerStyleFactory {
    /** The pool key is the combination of intrinsic values. */
    private record Key(String icon, String colour, int size) { }

    private final Map<Key, MarkerStyle> pool = new ConcurrentHashMap<>();

    MarkerStyle get(String icon, String colour, int size) {
        return pool.computeIfAbsent(new Key(icon, colour, size),
                k -> new MarkerStyle(k.icon(), k.colour(), k.size()));
    }
    // ... poolSize() ...
}

/** CONTEXT: one per marker. Holds extrinsic state + a REFERENCE to a shared style. */
record Marker(double lat, double lon, MarkerStyle style) {
    String draw() { return style.draw(lat, lon); }
}
```

`computeIfAbsent` makes get-or-create **atomic and thread-safe**; the `Key` record gives correct `equals`/`hashCode`.

```
  markers: 100000, styles in pool: 3
  icon memory without flyweight: ~200000 MB
  icon memory with flyweight:    ~6 MB
  Integer 127 == 127 -> true  (cached -128..127)
  Integer 128 == 128 -> false (outside default cache)
```

Full runnable code: code/lld/ch09/flyweight/ (see the Full Edition).

Flyweights already in Java

- `Integer.valueOf` (autoboxing) caches **−128..127** (JLS-required); 128 may differ (`-XX:AutoBoxCacheMax` can raise the bound). **Compare boxed numbers with `equals`.** `Short`, `Byte`, `Long` (same range), `Character` (0–127), `Boolean.TRUE/FALSE` likewise.
- **String pool** – literals and compile-time constants (`"hel" + "lo"`) are interned; `new String("hello")` is new; `intern()` returns the pooled one.
- **Enums** – one instance per constant.

Flyweight is not a cache

	Flyweight	Cache
Purpose	Make data *fit in memory*	Make repeated work *faster*
Contents	One copy of immutable state shared by many	Stored results of costly work
Remove entries?	Not while referenced	Any time
Effect of removal	Lost sharing / duplicates	Only a slower next call

Two mistakes that undo the pattern

- **A mutable flyweight** – one caller changes a colour and every marker changes.
- **An ever-growing pool keyed by open-ended input** – a leak disguised as an optimisation. Safe only for a **small, bounded** set of intrinsic values; otherwise you've built a cache (weak refs, eviction) – call it one.

Measure first

It trades readability for memory – without profiling it's premature optimisation. Answer with numbers: *how many objects at once?* and *how much of each is identical?*

Trade-offs

Pros	Cons
Memory can fall by orders of magnitude	State split across two places
Costly construction once per distinct value	The pool holds strong references (can leak)
CPU-cache friendly	**Thread safety is on you**
"Load once" logic in one home	Tempting too early; extrinsic state passed on every call

When not to use it

You haven't measured; the shared part can't be immutable; counts are small; unique state dominates; the key space is unbounded (that's a cache).

Common mistakes

Extrinsic state inside the flyweight; a public constructor; a non-thread-safe pool (`HashMap` check-then-put); comparing boxed `Integer`s or `String`s with `==`.

Other patterns it is confused with

Looks like	Difference
Cache	See table above
Singleton	One instance total vs one **per distinct intrinsic value**
Object pool	Lends **mutable** objects and takes them back; flyweights are immutable, shared, never returned
Prototype	Makes many objects by copying; Flyweight avoids making them

Real-world examples

`Integer.valueOf`, `Boolean.valueOf`, `String.intern()`, enums (Python interns small ints/strings too); glyphs in word processors (GoF's example); game meshes/textures with GPU instancing; map pins; `ZoneOffset.ofHours(5)` (cached 15-minute multiples).

In modern languages

Any immutable value object + memoising factory (Java record + `computeIfAbsent`, JS `Object.freeze` + `Map`, C++ `shared_ptr<const Style>`).

In the interview

> **30-second answer:** "Flyweight saves memory with huge numbers of similar objects. I split state into intrinsic – shared and invariant, like an icon – and extrinsic – per object, like a position, passed into methods. The intrinsic part is an immutable flyweight from a factory that pools one instance per distinct value. `Integer.valueOf` and the String pool are JDK examples. I'd only do it after measuring."

- *Q: Why is `Integer 128 == 128` false?* – A: `valueOf` only guarantees caching for –128..127; use `equals`.
- *Q: Thread-safe factory?* – A: `ConcurrentHashMap.computeIfAbsent` – creation runs at most once per key.

Self-check

1. **Q:** Chess piece: intrinsic vs extrinsic? **A:** Type, colour, image vs its square.
2. **Q:** What if `MarkerStyle` had `setColour`? **A:** Changing one marker changes every marker sharing it.
3. **Q:** When does a pool leak? **A:** When keys are unbounded (e.g. user input) and nothing evicts.

TL;DR

Share what repeats, pass in what varies. **Immutable** flyweight + **factory-owned**, thread-safe pool. `Integer.valueOf` (–128..127), String pool, enums. **Measure first**; not a cache.

9.7 Proxy Pattern

What it is

Proxy puts a **stand-in with the identical interface** in front of a real object to control **who** may use it, **when** it's created, or **how much** each use costs. GoF intent: *provide a surrogate or placeholder for another object to control access to it* (a.k.a. **Surrogate**). Analogy: a credit card standing in for your bank account.

Term	Ours
Subject	`Report`, `RateService`, `Device`
Real Subject	`RealReport`, `LiveRateService`, `RealDevice`
Proxy	`LazyReport`, `AuditedReport`, `CachingRateProxy`, `RemoteRateProxy`, `SharedDevice` handles

The problem it solves

Callers shouldn't touch the real object directly (or yet): it's **costly and usually unneeded** (a 100 MB report when most screens show only its title); **only some callers are permitted**; it's **on another machine**; **each call must be logged/counted** but the class isn't yours. Same interface → insert it **without editing callers** – which is also the danger: a slow call **looks cheap**.

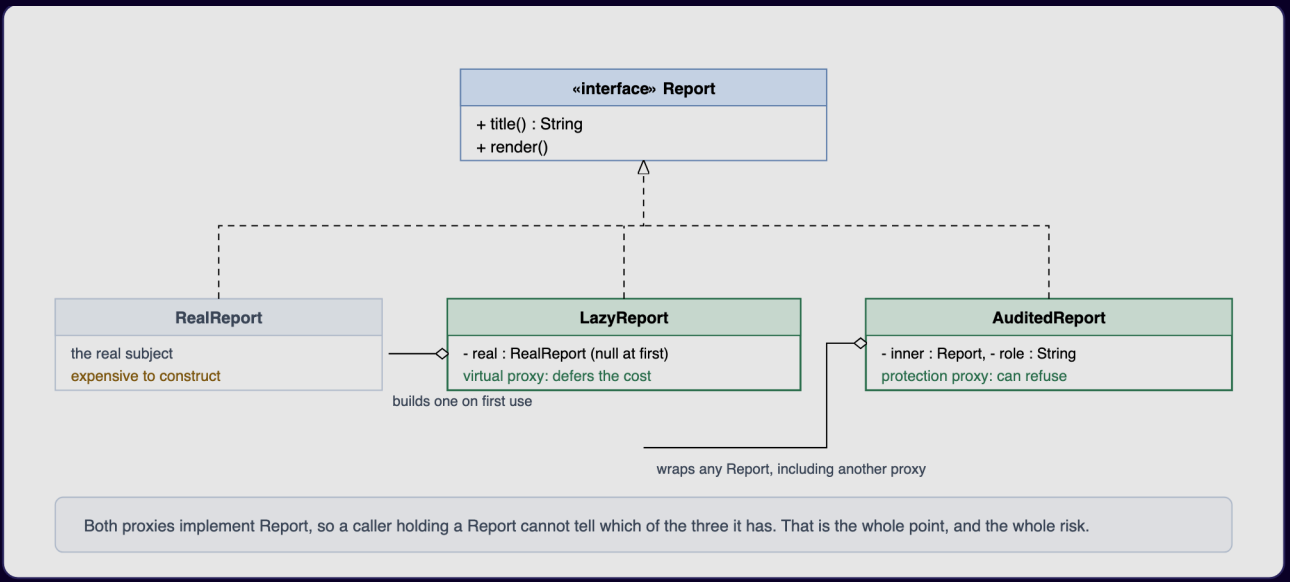


Figure – Proxy class diagram: `RealReport`, `LazyReport` (virtual proxy) and `AuditedReport` (protection proxy) all implement `Report`; `LazyReport` creates the `RealReport` itself, `AuditedReport` wraps any `Report` (course diagram)

How it works

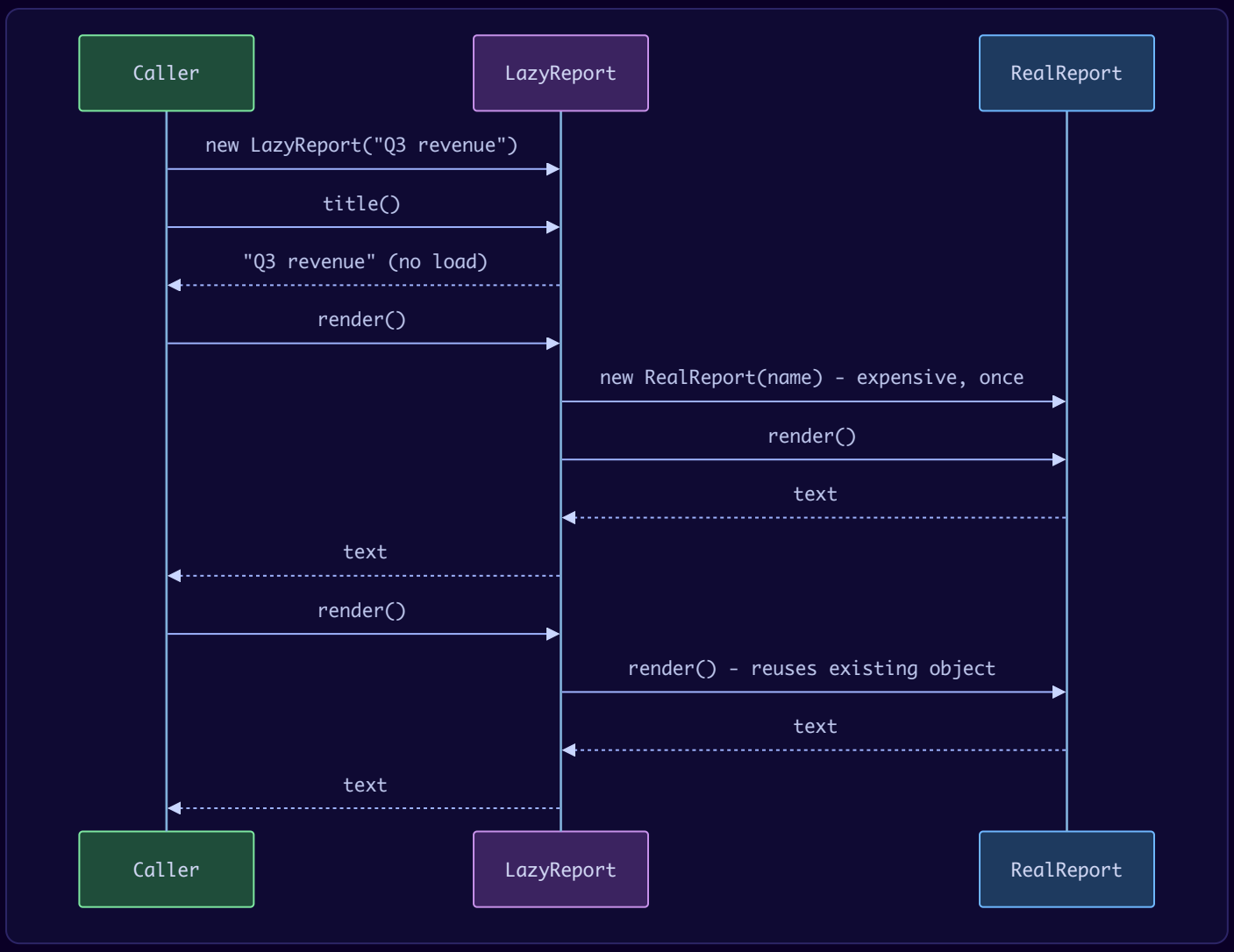
1. Define/reuse the **Subject** interface. 2. A proxy implementing it, holding the real subject **or what's needed to create it**. 3. In each method do the proxy's job (create lazily, check permission, check cache, marshal, count), then **delegate or refuse**. 4. Give callers the proxy (often via a factory or DI).

Kinds of proxy

Kind	Controls	Ours / real
Virtual	*When* it's created (lazy)	`LazyReport`; Hibernate lazy entities
Protection	*Who* may call	`AuditedReport`; `Collections.unmodifiableList`
Remote	*Where* it lives	`RemoteRateProxy` (stub) + `RateServer` (skeleton); RMI/gRPC
Caching	*Whether* the call is needed	`CachingRateProxy` (LRU + TTL)
Logging / monitoring	Records calls	`LoggingHandler` (dynamic proxy)
Smart reference	Bookkeeping: ref counting, locking, load on first use	`SharedDevice` closes on last handle

GoF names four (remote, virtual, protection, smart reference); caching and logging are later variants (GoF would file logging under smart reference).

Structure


Caller
LazyReport
RealReport
new LazyReport("Q3 revenue")
title()
"Q3 revenue" (no load)
render()
new RealReport(name) - expensive, once
render()
text
text
render()
render() - reuses existing object
text
text
Caller
LazyReport
RealReport

Java example

```
final class LazyReport implements Report {
    private final String name;
    private volatile RealReport real;      // created by the proxy itself, not handed in

    LazyReport(String name) { this.name = name; }

    @Override public String title() { return name; }   // no load needed

    @Override
    public String render() { return realReport().render(); }

    private RealReport realReport() {
        RealReport r = real;
        if (r == null) {
            synchronized (this) {
                r = real;
                if (r == null) real = r = new RealReport(name);
            }
        }
        return r;
    }
}

/** PROTECTION PROXY: checks permission, may REFUSE the call. */
final class AuditedReport implements Report {
    private final Report inner;
    private final Role callerRole;
    private final Set<Role> allowed;
    // ... constructor copies allowed with Set.copyOf ...
    @Override public String title() { return inner.title(); }  // titles are public

    @Override
    public String render() {
        if (!allowed.contains(callerRole)) {
            throw new SecurityException(callerRole + " may not render " + inner.title());
        }
        return inner.render();
    }
}
```

- **Thread-safe lazy init** – double-checked locking; the field **must be** `volatile`, or two threads may build twice (a real bug if construction opens a connection or charges a card).
- `AuditedReport` wraps a `LazyReport`: the viewer is refused **before anything loads**.
- `CachingRateProxy` is **bounded** (LRU via access-ordered `LinkedHashMap` + `removeEldestEntry`) and **time-limited** (TTL): 5 calls → 4 lookups. Rates are `BigDecimal`.
- `RemoteRateProxy` marshals `"rate|USD|INR"` over a simulated network to `RateServer`.
- `SharedDevice` hands out handles; the device opens on the first and closes on the last, all under one lock, `close()` idempotent.

Full runnable code: code/lld/ch09/proxy/ (see the Full Edition).

Static vs dynamic proxies

	Static	Dynamic
Made by	Hand-written per interface	Generated at **run time**
Code	One class each, every method	One `InvocationHandler` for **any** interface
Type safety	Full	Reflective (`Method` + `Object[]`)
Good for	Type-specific logic (lazy load, domain access rules)	**Cross-cutting**: logging, timing, transactions, security
API	–	`Proxy.newProxyInstance(loader, interfaces, handler)`
Limit	Boilerplate	JDK proxies are **interface-only**; classes need CGLIB/ByteBuddy

JDK mechanics: `newProxyInstance` generates `$Proxy0` implementing the interfaces → every call goes to `handler.invoke(proxy, method, args)` → the handler works and calls `method.invoke(target, args)` → **unwrap** `InvocationTargetException` (`e.getCause()`) so callers see the real exception.

The self-invocation trap (and Spring AOP)

In the demo, `convert` called `rate(...)` on `this`, bypassing the logging proxy. **Spring AOP** works the same way: `@Transactional`, `@Cacheable`, `@Async`, `@PreAuthorize` and `@Aspect` advice wrap your bean in a proxy – a **JDK dynamic proxy** if it implements an interface, else **CGLIB** (Spring Boot 2.0+ defaults to CGLIB). **Self-calls aren't intercepted** (`this.saveAudit()` with `REQUIRES_NEW` is silently ignored); with CGLIB, `final` classes can't be proxied and `final` / `private` methods can't be advised. Fixes: move the method to another bean; inject the bean's own proxy (lazy self-reference); or (discouraged) `AopContext.currentProxy()` with `exposeProxy = true`. AspectJ weaving avoids proxies altogether.

Proxy vs Decorator

GoF: similar implementations, different **purpose** – Decorator *adds responsibilities*, Proxy *controls access*.

	Proxy	Decorator
Purpose	**Controls access**	**Adds behaviour**
Call gets through?	Not necessarily – defer, cache, refuse	Normally yes
Who creates the real object?	**Often the proxy**	**The caller** hands it in
Client knows?	Usually not (factory/DI supplies it)	Usually yes (builds the stack)
Layers	Usually one	Often several

Creation test: `LazyReport(String name)` builds its subject; `LoggingRepository(UserRepository inner)` is given one. `AuditedReport` *is* handed its subject – only intent separates it from a decorator. A caching wrapper is honestly both; ask instead whether its cache is bounded and invalidated.

Where proxies go wrong

Unsynchronised lazy creation (duplicates); a caching proxy with no bound or expiry (leak + stale data); **concealed cost** – a remote proxy used in a loop (fix by making cost visible: batch APIs, `fetch...` names); **ORM N+1** – touching `order.getCustomer().getName()` for 100 orders fires 100 queries.

Trade-offs

Pros	Cons
Pay later for costly construction	Extra indirection in stack traces
No caller changes (Open/Closed)	**Latency and failures disguised** as local calls
Home for cross-cutting rules on classes you can't edit	Lazy creation raises thread-safety questions
Permission checks kept out of the object	Unbounded caches leak and go stale

When not to use it

The real object is cheap; callers need to see the cost (`fetchOverNetwork()`); the framework already generates proxies (ORM, DI, service mesh); you want to add behaviour (Decorator).

Common mistakes

Non-`volatile` DCL field; not unwrapping `InvocationTargetException`; expecting Spring annotations on self-calls, `private` or `final` methods; unbounded caching proxies; remote proxies in tight loops.

Real-world examples

ORM lazy loading (Hibernate `getReference`, lazy associations — the source of N+1 and `LazyInitializationException`); **Spring AOP** (and CGLIB-enhanced `@Configuration` classes unless `proxyBeanMethods = false`); RMI/gRPC/Thrift/SOAP stubs; `java.lang.reflect.Proxy` (Spring Data repositories, MyBatis mappers); `Collections.unmodifiableList`; Envoy/Istio sidecars, Nginx/Cloudflare reverse proxies; C++ `shared_ptr`.

In modern languages

JS built-in `Proxy`; Python `__getattr__`; C# `DispatchProxy` and `Lazy<T>`; C++ smart pointers.

Other patterns it is confused with

Looks like	Difference
Decorator	Adds behaviour to a handed-in object; Proxy controls access, often creates the subject
Adapter	**Different** interface on purpose; Proxy has the **same** one
Facade	New simpler interface over **several**; Proxy the same interface over **one**
A cache	A caching proxy *is* a cache — still needs bounds and eviction

In the interview

> **30-second answer:** "A Proxy is a same-interface stand-in that controls access. I name the kind: virtual defers creation, protection checks permissions, remote hides the network, caching reuses results, smart reference does bookkeeping like ref counting. Callers don't change — which is also the danger, since an expensive call looks cheap. Lazy creation needs thread safety, and ORM lazy proxies cause N+1. In Java I'd use `java.lang.reflect.Proxy` for cross-cutting concerns — that's how Spring AOP does `@Transactional`."

- *Q: Why doesn't `@Transactional` work on a same-class call?* — A: `this.method()` never goes through the proxy.
- *Q: JDK proxy vs CGLIB?* — A: JDK needs interfaces; CGLIB subclasses, so no `final` classes/methods.
- *Q: N+1 fix?* — A: Fetch joins, entity graphs or batch fetching.

Self-check

1. **Q:** Kinds of `LazyReport` and `AuditedReport`? **A:** Virtual; protection.
2. **Q:** Why wasn't the inner `rate()` logged? **A:** Self-invocation — `convert()` called `rate()` on `this`.
3. **Q:** Two must-haves for a caching proxy? **A:** A size bound and an expiry/invalidation policy.

TL;DR

Same interface, controlled access: virtual, protection, remote, caching, logging, smart reference. Static = hand-written; dynamic = `Proxy` + `InvocationHandler`. Spring AOP = proxies → beware self-invocation, `final`, `private`; watch thread safety and hidden costs.

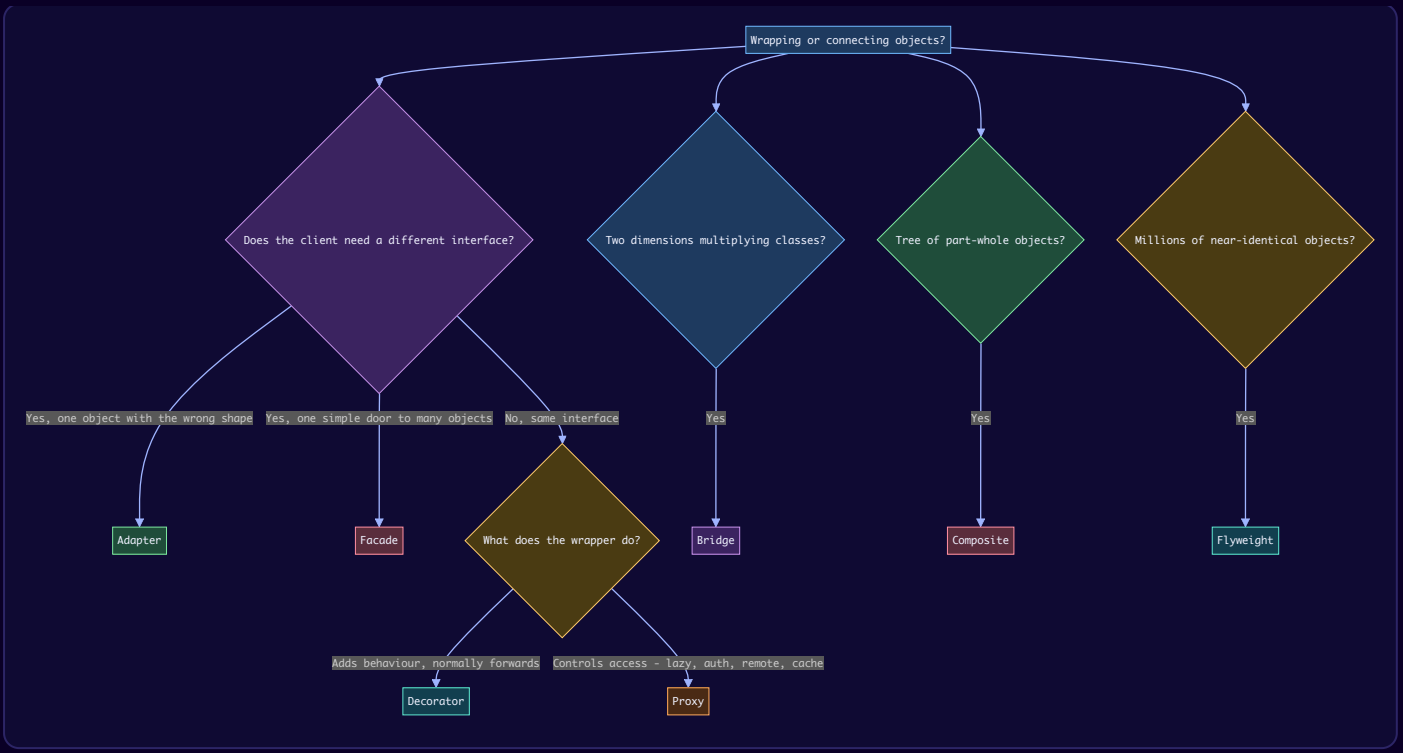
9.8 Putting it all together – comparison

Summary of all seven structural patterns

Pattern	Distinctive feature	Use it when	Example
Adapter	Bridges incompatible interfaces	An existing class doesn't fit	Plug adapter; `StripeGateway`
Bridge	Separates abstraction from implementation	Two dimensions vary independently	Shapes × renderers; notifications × channels
Composite	Part-whole tree, one interface	Treat single and group uniformly	Drawing groups; file system
Decorator	Adds responsibilities dynamically	Run-time combinations of behaviour	Scroll/border on a view; Java I/O
Facade	Simple interface to a subsystem	Callers repeat one common path	"Watch movie" button; `OrderFacade`
Flyweight	Shares common state	Huge numbers of similar objects	Word-processor glyphs; map markers
Proxy	Placeholder controlling access	Lazy, auth, remote, cache, logging, ref counting	Proxy server; Hibernate lazy entity

Adapter vs Decorator vs Proxy vs Facade vs Bridge

	Adapter	Decorator	Proxy	Facade	Bridge
Intent	Convert an interface	Add behaviour, same interface	Control access	Simplify a subsystem	Vary abstraction and implementation independently
Client sees	**Different** (yours)	**Same**	**Same**	**New, simpler**	Abstraction's own
Behaviour	Unchanged (translated)	**Extended**	**Controlled**	Unchanged, fewer calls	Delegated to implementor
Wraps	One	One (stackable)	One	**Several**	One implementor
Creates wrapped?	Passed in	Passed in by client	**Often itself**	Passed in / owned	Passed in
Introduced	**After the fact**	When a concern is added	When access needs control	When callers repeat a sequence	**Up front**
Trigger	"Vendor API doesn't match"	"Sometimes logging, sometimes caching"	"Expensive / remote / restricted"	"Every caller copies 5 steps"	"Two-noun class names"
Example	`Arrays.asList`, `InputStreamReader`	`BufferedInputStream`, `synchronizedList`	`reflect.Proxy`, `@Transactional`, Hibernate	SLF4J, service layers	JDBC + drivers, AWT peers



Memory hooks

Adapter – make it fit · **Decorator** – add to it · **Proxy** – guard it · **Facade** – simplify it · **Bridge** – split it · **Composite** – nest it · **Flyweight** – share it.

Chapter self-check

1. **Q:** Which wrap exactly one object? **A:** Adapter, Decorator, Proxy (Bridge holds one implementor by design). Facade wraps several; Composite holds many.
2. **Q:** Same code shape, different intent/timing? **A:** Adapter vs Bridge (after vs up front); Bridge vs Strategy; Decorator vs Proxy.
3. **Q:** `WindowsButton`, `MacButton`, `WindowsCheckbox` … and Linux coming? **A:** Bridge (widget × platform), often with Abstract Factory choosing the implementor.
4. **Q:** Why is `Integer.valueOf(100) == Integer.valueOf(100)` true? **A:** Flyweight (Integer cache).

Chapter TL;DR

- Structural patterns **compose objects into bigger structures** while staying flexible.
- **Adapter / Decorator / Proxy / Facade** share a shape – tell them apart by **what changes as the call passes** and **how many objects are wrapped**.
- **Bridge** stops two-dimension class explosion; **Composite** makes trees uniform; **Flyweight** shares immutable state.
- In interviews: name the pattern, the **constraint** justifying it, its **trap** (vendor leaks, wrong split, cycles, decorator order, god facade, mutable flyweight, self-invocation) and a **real JDK/Spring example**.